Synoptic Question 5: Geostationary Communications Satellite

Reading Passage

Read the following article and then answer the questions that follow.

Modern global communications rely heavily on a network of artificial satellites orbiting the Earth. One of the most important types is the geostationary satellite. A satellite in a geostationary orbit remains at a fixed point above the Earth's equator. This is achieved by matching the satellite's orbital period exactly to the Earth's rotational period. This fixed position allows a ground-based antenna to be pointed permanently at the satellite, making it ideal for continuous communication services like satellite television and long-distance telephone calls.

To achieve this orbit, the satellite must be placed at a very specific altitude where the gravitational force provides the exact centripetal force required for it to travel at the necessary speed. The signals transmitted to and from the satellite are high-frequency microwaves, which travel in straight lines and can pass through the Earth's ionosphere with minimal distortion.

A typical communications satellite is powered by large solar panel arrays which convert sunlight into electrical energy. This energy is used to power the onboard electronics and to transmit signals back to Earth. The power of the transmitted signal decreases with distance according to the inverse square law. As the signal passes through the atmosphere, it also experiences some loss of strength, known as attenuation.

Data:

Table 1: Data for Earth and a typical Satellite

Parameter	**Value**
Mass of Earth, M_E	5.97×10^{24} kg
Radius of Earth, R_E	6.37×10^{6} m
Gravitational constant, G	6.67×10^{-11} N m^2 kg^{-2}
Mass of satellite, m	4500 kg
Frequency of uplink signal	14.0 GHz
Power of transmitted signal from satellite, P_{trans}	2.50 kW
Area of satellite's solar panels	50 m^2
Efficiency of solar panels	28%
Solar radiation intensity at orbit, I_{solar}	1360 W m^{-2}

Questions

(a) For a satellite to be in a geostationary orbit:

(i) State the period of the orbit in seconds.

(ii) State the direction of the orbit relative to the Earth's rotation.

(b) By equating the gravitational force on the satellite with the required centripetal force, show that the radius r of the satellite's orbit is given by the expression:

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

where T is the period of the orbit.

(c) Calculate:

(i) The radius of the geostationary orbit.

(ii) The altitude of the satellite above the Earth's surface.

(iii) The linear speed of the satellite.

(d) The signals used for communication are microwaves.

(i) A 14.0 GHz signal is sent from a ground station to the satellite (the uplink). Calculate the wavelength of this signal.

(ii) Calculate the minimum time delay for a signal to travel from a ground station on the equator to the satellite and back to the ground station.

(e) The satellite transmits a signal with a power of 2.50 kW. This signal spreads out uniformly in a sphere.

(i) Assuming the signal spreads isotropically (uniformly in all directions), calculate the intensity of the signal when it reaches the Earth's surface.

(ii) The signal is received by a parabolic dish on Earth with a diameter of 1.2 m. Calculate the power received by this dish.

(f) The satellite is powered by solar panels.

(i) Calculate the total electrical power generated by the solar panels.

(ii) Suggest why the power generated might be greater than the power required to transmit the signal.

(g) Calculate the gravitational potential energy of the satellite in its orbit. (Gravitational potential at a point is defined as zero at infinity).

Solutions

(a) (i) The period must be the same as the Earth's rotational period, which is 24 hours. $T = 24\text{ h} \times 60\text{ min/h} \times 60\text{ s/min} = 86400\text{ s}$.

(ii) The direction of the orbit must be the same as the Earth's rotation (from west to east).

(b) Gravitational Force = Centripetal Force

$$\frac{GM_E m}{r^2} = mr\omega^2$$

Since angular velocity $\omega = \frac{2\pi}{T}$:

$$\frac{GM_E}{r^2} = r\left(\frac{2\pi}{T}\right)^2 = \frac{4\pi^2 r}{T^2}$$

Rearranging for r^3:

$$r^3 = \frac{GM_E T^2}{4\pi^2}$$

$$r = \sqrt[3]{\frac{GM_E T^2}{4\pi^2}}$$

(c) (i) $r = \sqrt[3]{\frac{(6.67\times10^{-11})(5.97\times10^{24})(86400)^2}{4\pi^2}} = \sqrt[3]{7.53\times10^{22}}$
$r = 4.22\times10^7$ m.

(ii) Altitude $h = r - R_E = (4.22\times10^7) - (6.37\times10^6) = 3.58\times10^7$ m (or 35,800 km).

(iii) Linear speed $v = r\omega = r\frac{2\pi}{T} = (4.22\times10^7)\frac{2\pi}{86400} = 3070\text{ m s}^{-1}$.

(d) (i) $f = 14.0\text{ GHz} = 14.0\times10^9$ Hz. Speed of light $c = 3.00\times10^8\text{ m s}^{-1}$.
Wavelength $\lambda = \frac{c}{f} = \frac{3.00\times10^8}{14.0\times10^9} = 0.0214$ m.

(ii) The signal travels from ground to satellite and back, covering a total distance of $2h$.
Total distance $= 2\times(3.58\times10^7) = 7.16\times10^7$ m.
Time delay $t = \frac{\text{distance}}{\text{speed}} = \frac{7.16\times10^7}{3.00\times10^8} = 0.239$ s.

(e) (i) The signal power spreads out over the surface area of a sphere with radius equal to the satellite's altitude.
Surface area $A = 4\pi h^2 = 4\pi(3.58\times10^7)^2 = 1.61\times10^{16}\text{ m}^2$.
Intensity $I = \frac{P_{trans}}{A} = \frac{2500}{1.61\times10^{16}} = 1.55\times10^{-13}\text{ W m}^{-2}$.

(ii) Radius of dish $r_{dish} = 0.6$ m. Area of dish $A_{dish} = \pi r_{dish}^2 = \pi(0.6)^2 = 1.13\text{ m}^2$.
Power received $= I\times A_{dish} = (1.55\times10^{-13})\times1.13 = 1.75\times10^{-13}$ W.

(f) (i) Power incident on panels $= I_{solar}\times\text{Area} = 1360\times50 = 68000$ W.
Electrical power generated = Power incident × Efficiency $= 68000\times0.28 = 19040$ W $=$ 19.0 kW.

(ii) The extra power is needed to run all the satellite's onboard electronic systems (computers, sensors, cooling systems) and to charge batteries for times when the satellite might be in the Earth's shadow (eclipse).

(g) Gravitational potential energy $U = -\frac{GM_E m}{r}$.

$$U = -\frac{(6.67\times10^{-11})(5.97\times10^{24})(4500)}{4.22\times10^7}$$

$$U = -4.24\times10^{10}\text{ J}$$

[Diagram of a satellite in geostationary orbit above the Earth's equator]

Figure 1: Geostationary Satellite Orbit

Cambridge IGCSE Physics (0625) – Paper 4

Extended Theory – Variant 2 (Higher Difficulty)

Appendix: Open-ended Conceptual Questions, Sample Solutions and Mark Scheme

Instructions

- Answer all questions.
- Write clear explanations, using physical principles and equations where appropriate.
- The number of marks is given in brackets [ ] at the end of each question or part question.

Question Paper (Variant 2)

Question 1: Orbital Dynamics and Energy

A satellite of mass 750 kg is moved from a circular orbit of radius $r_1 = 7.0 \times 10^6$ m to a higher circular orbit of radius $r_2 = 1.4 \times 10^7$ m. Take $G = 6.67 \times 10^{-11}$ N m^{2}kg^{-2} and Earth mass $M = 6.0 \times 10^{24}$ kg.

(a) Explain why the gravitational force on the satellite decreases with increasing orbital radius, and discuss how this affects orbital speed. [4]

(b) Calculate the orbital speed in the initial orbit and in the higher orbit. [4]

(c) Determine the change in total mechanical energy of the satellite-Earth system when transferring from r_1 to r_2. State and interpret the sign of your answer. [6]

Question 2: Thermal Physics and Experimental Design

A student investigates the cooling of hot water in identical beakers placed in different environments.

(a) Describe how the rate of cooling depends on the temperature difference between water and surroundings. Refer to particle-level and energy-transfer arguments. [4]

(b) Design an experiment to compare cooling rates in still air and under forced convection (fan). Include:

- apparatus,
- variables to control,
- measurements,
- method to improve reliability.

[6]

(c) A 0.20 kg sample of water cools from 80 °C to 65 °C in 5.0 min. Estimate the average rate of thermal energy loss. $c_{\text{water}} = 4200\,\text{Jkg}^{-1}\text{K}^{-1}$. [4]

Question 3: Electricity, Internal Resistance and Modelling

A battery of e.m.f. 12.0 V and internal resistance 0.80 Ω is connected to a variable load resistor R.

(a) Explain physically why terminal potential difference changes when current changes in a source with internal resistance. [4]

(b) Derive an expression for terminal p.d. V in terms of $\mathcal{E}$, I, and internal resistance r. [3]

(c) For $R = 5.2\,\Omega$, calculate the current and terminal p.d. [4]

(d) Discuss how you would use a V-I graph to determine both e.m.f. and internal resistance experimentally. [4]

Question 4: Waves, Interference and Evaluation

Two coherent sound sources emit in phase at frequency 680 Hz. Speed of sound in air is $340\,\mathrm{ms}^{-1}$.

(a) Define coherence and explain why it is required for a stable interference pattern. [3]

(b) Calculate the wavelength of the sound. [2]

(c) At a point P, the path difference from the two sources is 0.75 m. Determine whether interference at P is constructive or destructive. Justify clearly. [4]

(d) A listener reports that nodes are less distinct in a noisy hall than in a quiet lab. Explain this observation in terms of superposition and real-world limitations. [5]

Sample Solutions (Variant 2)

Q1 Sample Solution

(a) $F = \dfrac{GMm}{r^2}$, so increasing r reduces force by inverse-square law. For circular motion, gravity provides centripetal force; weaker gravity at larger r means smaller centripetal acceleration and lower orbital speed.

(b) For circular orbit, $v = \sqrt{\dfrac{GM}{r}}$.

$$v_1 = \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{7.0 \times 10^6}} \approx 7.56 \times 10^3 \text{ m s}^{-1}$$

$$v_2 = \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{1.4 \times 10^7}} \approx 5.35 \times 10^3 \text{ m s}^{-1}$$

(c) Total energy in circular orbit: $E = -\dfrac{GMm}{2r}$.

$$E_1 = -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(750)}{2(7.0 \times 10^6)} \approx -2.14 \times 10^{10} \text{ J}$$

$$E_2 = -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(750)}{2(1.4 \times 10^7)} \approx -1.07 \times 10^{10} \text{ J}$$

$$\Delta E = E_2 - E_1 \approx +1.07 \times 10^{10} \text{ J}$$

Positive change means energy must be supplied to raise the satellite to the higher orbit.

Q2 Sample Solution

(a) Larger temperature difference gives a larger net rate of energy transfer to surroundings (conduction/convection/radiation). At particle level, faster water molecules transfer energy more rapidly to cooler surroundings.

(b) Example method: two identical beakers, same mass of water, same initial temperature, thermometers/data logger, stopwatch, fan. One beaker in still air; one with fixed fan speed and fixed distance. Control mass of water, container, starting temperature, room conditions (as far as possible). Record temperature at fixed intervals (e.g. every 30 s for 10 min), repeat trials, average results, plot T-time curves and compare gradients.

(c)

$$Q = mc\Delta T = (0.20)(4200)(15) = 12600 \text{ J}$$

$$t = 5.0 \text{ min} = 300 \text{ s}, \quad P_{\text{avg}} = \frac{Q}{t} = \frac{12600}{300} = 42 \text{ W}$$

Q3 Sample Solution

(a) Internal resistance causes a voltage drop Ir inside the source when current flows. As current increases, this internal drop increases, so terminal p.d. available to external circuit decreases.

(b)

$$\mathcal{E} = V + Ir \Rightarrow V = \mathcal{E} - Ir$$

(c) Total resistance $= R + r = 5.2 + 0.8 = 6.0\,\Omega$.

$$I = \frac{\mathcal{E}}{R + r} = \frac{12.0}{6.0} = 2.0 \text{ A}$$

$$V = IR = (2.0)(5.2) = 10.4 \text{ V}$$

(equivalently $V = \mathcal{E} - Ir = 12.0 - (2.0)(0.8) = 10.4\,\text{V}$).

(d) Measure V and I for different loads and plot V (y-axis) against I (x-axis). From $V = \mathcal{E} - Ir$: y-intercept gives $\mathcal{E}$, gradient $= -r$, magnitude of gradient gives internal resistance.

Q4 Sample Solution

(a) Coherent sources have constant phase difference and same frequency. Without this, maxima/minima positions fluctuate and stable interference is not observed.

(b)

$$\lambda = \frac{v}{f} = \frac{340}{680} = 0.50\,\mathrm{m}$$

(c)

$$\frac{\Delta x}{\lambda} = \frac{0.75}{0.50} = 1.5 = \frac{3}{2}$$

Half-integer multiple of λ gives destructive interference (minimum intensity).

(d) Background reflections/noise introduce additional waves and random phases, reducing contrast between maxima and minima. Finite source size and imperfect coherence also blur nodal positions.

Mark Scheme (CIE P4 Style – Variant 2 Only)

Q1 (14 marks)

- (a) Inverse-square force relation stated; force decreases with radius; gravity provides centripetal force; lower force implies lower orbital speed. [4]
- (b) Correct use of $v = \sqrt{GM/r}$; $v_1 \approx 7.56 \times 10^3\,\mathrm{m\ s^{-1}}$; $v_2 \approx 5.35 \times 10^3\,\mathrm{m\ s^{-1}}$. [4]
- (c) Uses $E = -GMm/(2r)$; evaluates E_1 and E_2; finds $\Delta E \approx +1.07 \times 10^{10}\,\mathrm{J}$; correct sign interpretation (energy input required). [6]

Q2 (14 marks)

- (a) Cooling rate linked to temperature difference; valid energy-transfer reasoning; particle/process explanation. [4]
- (b) Suitable apparatus and method; clear control variables; interval measurements/table/graph; repeats and averaging for reliability. [6]
- (c) $Q = mc\Delta T$ correct; unit conversion to seconds; $P = Q/t$ and answer 42 W. [4]

Q3 (15 marks)

- (a) Internal p.d. loss explained as Ir; terminal p.d. decreases as current increases. [4]
- (b) Correct derivation $V = \mathcal{E} - Ir$. [3]
- (c) Total resistance, current, and terminal p.d. all correct with units. [4]
- (d) Correct graph method, intercept and gradient interpretation, and extraction of $\mathcal{E}$ and r. [4]

Q4 (14 marks)

- (a) Coherence definition includes constant phase difference and same frequency; reason for stable pattern. [3]
- (b) Correct wavelength 0.50 m. [2]
- (c) Correct condition test using path difference; identifies destructive interference with justification. [4]
- (d) Evaluative explanation using superposition, noise/reflections, and coherence limitations. [5]

Total: 57 marks